# Group Coursework Submission Form

## Specialist Masters Programme

| | |
|---|---|
| **Please list all names of group members:**<br>(Surname, first name)<br>1. Remizov, Artem<br>2. Shi, Heqing<br>3. Tsoy, Alexandr<br>4. Zhao, Jiahua | **GROUP NUMBER:** **6** |
| **MSc in:** Financial Mathematics | |
| **Module Code:** SMM306 | |
| **Module Title:** Advanced Stochastic Modelling | |
| **Lecturer:** Dr. Aleš Černý | **Submission Date:** 3 April 2020 |

**Declaration:**

By submitting this work, we declare that this work is entirely our own except those parts duly identified and referenced in my submission. It complies with any specified word limits and the requirements and regulations detailed in the coursework instructions and any other relevant programme and module documentation. In submitting this work we acknowledge that we have read and understood the regulations and code regarding academic misconduct, including that relating to plagiarism, as specified in the Programme Handbook. We also acknowledge that this work will be subject to a variety of checks for academic misconduct.

We acknowledge that work submitted late without a granted extension will be subject to penalties, as outlined in the Programme Handbook. Penalties will be applied for a maximum of five days lateness, after which a mark of zero will be awarded.

**Marker's Comments (if not being marked on-line):**

**Deduction for Late Submission of assignment:**

***For Students:***
**Once marked please refer to Moodle for your final coursework grade, including your Peer Assessment grade.**

The present report discusses the use of Finite Difference method in solving Večeř's PDE to price discretely monitored Asian put options. The following inputs are used: the spot stock price $S_0 = 100$; the strike matrix $\boldsymbol{K} = [95 \;\; 100 \;\; 105]$; the risk-free rate $r = 0.1$; the continuous dividend yield $\widehat{\delta} = 0.01$; the annualised volatility of the stock's log-return $\sigma = 0.3$; the options expire in 2 years. The Asian options are monitored weekly, i.e. $\Delta_t = \frac{1}{50}$.

## Question 1

Asian option's terminal payoff depends on the average of the underlying stock price over the life period of the option. For example, the payoff function of an Asian call option is:

$$\text{Payoff} = (\bar{S}_T - K_1 S_T - K_2)^+$$

where $\bar{S}_T$ is the stock price average, $K_1$ is the floating strike and $K_2$ is the fixed strike. As in our case, the option has fixed strike, therefore, we have $K_1 = 0$ and $K_2 = \boldsymbol{K}$. To price a fixed strike Asian put option with payoff function $(K_2 - \bar{S}_T)^+$, we can first price a call option, and then obtain the price of a put by the put-call parity.

*Proof.* Price of Asian put is derived from the put-call parity:

$$\begin{aligned} C_0 - P_0 &= e^{-rT}\mathbb{E}(\bar{S}_T - K_2)^+ - e^{-rT}\mathbb{E}(K_2 - \bar{S}_T)^+ \\ &= e^{-rT}\mathbb{E}(\bar{S}_T - K_2) \qquad \forall \bar{S}_T \\ \implies \qquad P_0 &= C_0 - e^{-rT}\mathbb{E}(\bar{S}_T - K_2) \qquad \forall \bar{S}_T \end{aligned} \tag{1}$$

■

Under the risk-neural measure $\mathbb{Q}$, the dynamics of the stock price is

$$dS_t = S_t\left(\left(r - \widehat{\delta}\right)dt + \sigma dW_t\right)$$

We know that an Asian option's payoff depends on

$$\bar{S}_T = \int_0^T S_t d\mu_t$$

where $\mu_t$ is a weighting factor depending on whether continuous or discrete monitoring is used. In our discrete setting, we have

$$d\mu_t = \frac{1}{n}\sum_{k=1}^{n} \delta\left(\frac{k}{n}T\right)dt$$

where $n$ is the number of monitored stock prices, including the initial price, and $\delta(.)$ denotes the Dirac delta function. Thus, in discrete time, the terminal payoff of a fixed strike Asian call option can be represented as

$$\int_0^T S_t d\mu t - K_2 \tag{2}$$

To price such an option, we need to find a self-financing portfolio $X_t$ such that it replicates the payoff in (2). According to Večeř (2002), there is a trading strategy $q_t$ which is the number of shares held at time $t$,

$$\begin{aligned} q_t &= e^{-\int_t^T d\widehat{\delta}_t} \cdot \int_t^T e^{-r(T-s)+\int_s^T d\widehat{\delta}_u} d\mu_s \\ &= e^{-\widehat{\delta}(T-t)}\frac{1}{n}\sum_{k=\lfloor\frac{nt}{T}\rfloor+1}^{n} e^{-\left(r-\widehat{\delta}\right)\left(\frac{(n-k)}{n}\right)T} \end{aligned}$$

and a self-financing portfolio $X_t$ that replicates the payoff of an Asian call option in (2), with the dynamics

$$\begin{aligned} dX_t &= q_t dS_t + r(X_t - q_t S_t)dt + q_t \widehat{\delta} S_t dt \\ &= rX_t dt + q_t \left( dS_t - rS_t dt + \widehat{\delta} S_t d_t \right) \end{aligned}$$

with initial value $X_0 = q_0 S_0 - e^{-rT} K_2$, and terminal value $X_T = \bar{S}_T - K_2$, and thus has the same value as an Asian call option.

Solution can be found through the SDE of $X_t$. However, multiple variables in this SDE cause some difficulties. To reduce the dimensionality of the process of $X_t$, we can set $e^{\widehat{\delta} t} S_t$ as a numéraire. Therefore, by changing the numéraire, a simpler process $Z_T$ can be obtained

$$Z_t = \frac{X_t}{e^{\widehat{\delta} t} S_t} \tag{3}$$

Let $f(t, S_t) = e^{-\widehat{\delta} t} S_t^{-1}$, and apply two-dimentional Itô's lemma to $Z_t = X_t \cdot f_t$, we have

$$\begin{aligned} df_t &= f\left(\sigma^2 - r\right) dt - \sigma f dW_t \\ dZ_t &= f_t dX_t + X_t df_t + d[X, f]_t \\ &= \left( Z_t - e^{-\widehat{\delta} t} \right) \sigma^2 dt - \left( Z_t - q_t e^{-\widehat{\delta} t} \right) \sigma dW_t \end{aligned}$$

To price the Asian call option using $Z_t$ under the no-arbitrage principle, we need to find a new measure under which $Z_t$ is a martingale. Thus, by measure changing technique, we have

$$dZ_t = -\left( Z_t - e^{-\widehat{\delta} t} q_t \right) \sigma d\widetilde{W}_t$$

where $\widetilde{W}_t = -\sigma t + W_t$. By changing our numéraire, and switching to the $\widetilde{\mathbb{P}}$-measure, we arrived at an equivalent martingale measure under which the process $Z_t$ is a martingale, and therefore, the pricing of the Asian option can be done properly under this risk-neutral measure using $Z_t$.

Since $X_t$ replicates the payoff of an Asian call option, the value of a fixed strike Asian call option at time 0 will be

$$\begin{aligned} V_0 &= e^{-rT} \mathbb{E}\left(X_T | \mathcal{F}_0\right)^+ \\ &= e^{-rT} e^{\widehat{\delta} T} S_T \widetilde{\mathbb{E}}\left(Z_T | \mathcal{F}_0\right)^+ \\ &= e^{-(r-\widehat{\delta})T} S_T \widetilde{\mathbb{E}}\left(Z_T | \mathcal{F}_0\right)^+ \\ &= S_0 \widetilde{\mathbb{E}}\left(Z_T | \mathcal{F}_0\right)^+ \end{aligned} \tag{4}$$

Since we know that $Z_t$ is Markovian, it can be shown that the conditional expectation of $Z_T$ given the filtration $\mathcal{F}_t$, i.e. $\widetilde{\mathbb{E}}\left(Z_T | \mathcal{F}_t\right)^+$ can be presented as a two-variable function $u(t, Z_t)$ which satisfies the following PDE

$$\begin{cases} \dfrac{\partial u}{\partial t} + \dfrac{1}{2} \left( Z_t - e^{-\widehat{\delta} t} q_t \right)^2 \sigma^2 \dfrac{\partial^2 u}{\partial Z_t^2} = 0 \\ u\left(T, Z_T\right) = \left(Z_T\right)^+ \end{cases} \tag{5}$$

and the price of such an Asian call option is $u\left(0, Z_0\right) S_0$. Therefore, our original pricing problem can be transformed into solving Večeř's PDE with its terminal condition as described in (5). This is one of the desired properties of Večeř's PDE: it is a simple and numerically stable one-dimentional PDE, it has zero drift, and it gives a unified approach to pricing both fixed and floating strike Asian options.

The closed form solution to Večeř's' PDE is difficult to derive. Instead, we use the Explicit Finite Differ-

ence method to approximate a numerical solution. By explicit finite difference scheme, the first step is to construct a reasonable uniform spatial grid for our state variable $Z_t$ and time variable $t$. To seek a better numerical result from our spatial grid, we need to curtail the range of $Z_t$ to a finite and meaningful range. According to the definition of $Z$ in (3), if we assume zero fixed strike, then $Z_{max} = 1$. And if we assume the fixed strike is double of the stock price, then $Z_{min} = -1$. This gives a narrowed and reasonable range of the state variable $Z$ in the spatial grid, $Z \in [-1, 1]$. As a starting point of our MATLAB implementation, we set the step size of $Z$ be $h = 0.1$,

$$Z_i = Z_{min} + i \cdot h = -1 + i \cdot 0.1 \qquad \forall\, i \in \{0, 1, 2, 3, ..., 20\}$$

and according to the monitoring frequency of our particular Asian option, $\Delta t = \frac{1}{50}$, we constructed our initial time variable grid with step size dt $= \Delta_t$ as

$$t_j = j \cdot \Delta_t = j \cdot \frac{1}{50} \qquad \forall\, j \in \{0, 1, 2, 3, ..., 100\}$$

Having decided the range of $Z_t$ and $t$, the next step is to set up our terminal and boundary conditions. From (5), the terminal condition at time $T$ is simply $(Z)^+$, i.e. we keep positive $Z_i$ and set negative $Z_i$ to zero. Consider a deep-in-the-money Asian call option, at time $T$, we will have $u(T, Z_T)_{max} = (Z_T)^+$ which equals to $Z_{20}$ in our grid. Thus the upper bound of $u(t, Z_t)$ is $Z_{20} = 1\ \forall\ t$. Similarly, $u(T, Z_T) = 0$ for a deep-out-of-money option, which corresponds to a lower bound of $u(t, Z_t) = 0\ \forall\ t$.

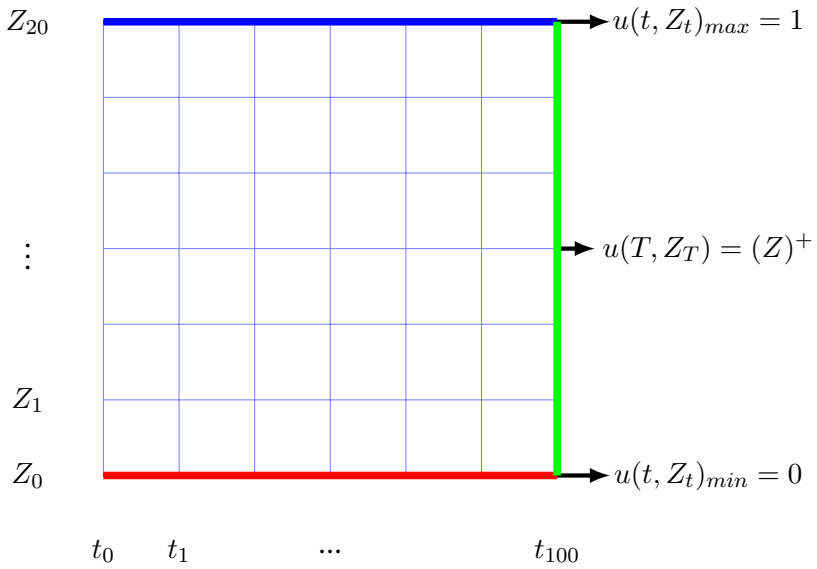


Figure 1: Our Uniformly Spaced Grid

Figure 1 illustrates the construction of our grid. The red line denotes our lower bound condition, the blue line denotes our upper bound condition, and the green line denotes our terminal condition. As in our case, the diffusion term in (5) is time-dependent, which means the transition probabilities used to approximate $u(t, Z_t)$ will also be time-dependent. By looping the terminal condition $u(T, Z_T)$ back to time zero, with the time-dependent transition probabilities, we obtained numerical solutions of $u(0, Z_0)$. Thus, according to (4), we know that the price of an Asian call option will be $S_0 u(0, Z_0)$. The Asian put price from (1) can be simplified further by noticing that $e^{-rT}\mathbb{E}(\bar{S}_T - K_2) = X_0$. Finally, we can calculate the price of the Asian put option. Our MATLAB implementation returns a price matrix $\boldsymbol{P} = [3.3686, 4.8662, 6.7045]$ for each strike in matrix $\boldsymbol{K}$, respectively.

## Question 2

For examining the convergence pattern, we need to find a benchmark price against which we will carry out our comparisons. As there is no closed-form solution for arithmetic average Asian option, we decided to use the Monte Carlo simulation (MCS) price as an approximation of the true price. However, the Monte Carlo price tends to show a substantial degree randomness and impreciseness. Following the approach explained by Kyriakou (2020), we have reduced the variance of Monte Carlo price through implementing the control variates technique. It exploits the fact that there is high correlation between arithmetic and geometric average Asian option prices. The latter has a closed-form solution derived by Turnbull and Wakeman (1991). Thus, for each simulation *j* we calculate the output of the geometric average put option (denoted as $P^{(j)}$) in addition to the already calculated payoff of the arithmetic average put option (denoted as $C^{(j)}$).

The Control Variates method relies on the following assumptions:

1) Pairs $[C^{(j)}, P^{(j)}]$ are i.i.d for all simulations *j* (valid by construction of MCS)
2) $C^{(j)}$ and $P^{(j)}$ are correlated (this is indeed the case for geometric and arithmetic Asian options)
3) $E(P^{(j)}) = \mu_P$ is known (this represents the true price of the geometric average option)

We run an auxiliary MCS (function `AsianMC_CtrlVar_coef`) to estimate the optimal coefficient $b^*$ that minimizes the variance of our main MC estimator. Auxiliary MCS needs to be implemented in order to prevent bias in our estimation process as the $b^*$ coefficient should be estimated based on an independent set of MC simulations.

$$\hat{b}^* = \frac{\mathrm{Cov}(C, P)}{\sigma_P^2}$$

Finally, in the main MCS (function `AsianMC_CtrlVar`) we construct an auxiliary process $C_b^{(j)}$, with help of which we derive the final estimator for arithmetic Asian put price:

$$C_b^{(j)} = C^{(j)} - b^* \left( P^{(j)} - \mu_P \right)$$

By running 1 million simulations using such technique, we achieved fairly precise estimates for the true price of an arithmetic average Asian put. The 95% confidence interval (CI) is reported in Table 1 on the next page. It can be seen that we are confident up to the second decimal place and the difference between the lower and upper bounds is no more than 0.002.

We explored the convergence of our Explicit FD price estimate by subdividing both the spatial and the time steps by *j* and $j^2$, where $j = 1, 2, ..., 10$ correspondingly. The results can be seen in the following figure (red dashed line represents the 95% MCS confidence interval):

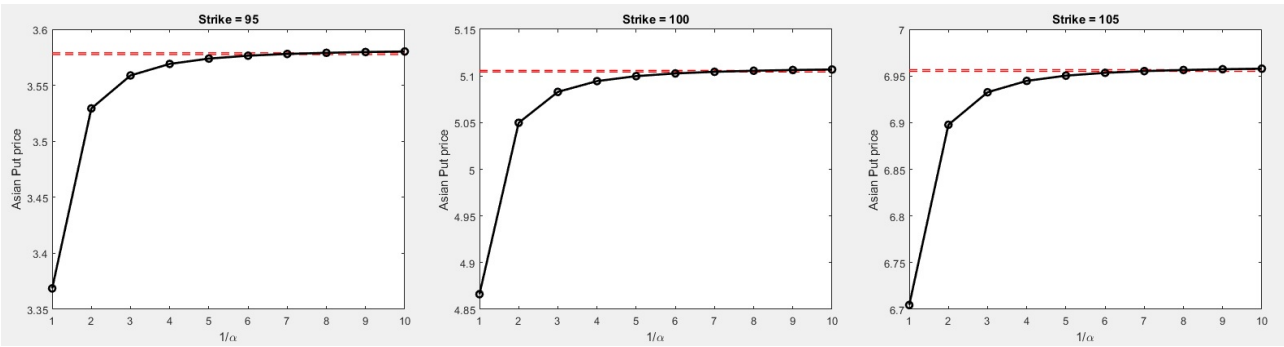


Figure 2: Convergence patterns of the Asian option price, with Strike = 95, 100, 105 respectively

Because of the stability of Večeř's PDE, we did not observe an oscillation of the Explicit FD price around the Monte Carlo price, thus no remedy was required. Table 1 reports our best estimates of Explicit FD prices. It can be seen that they are close to the Monte Carlo true price, with accuracy of up to $10^{-2}$ or $10^{-3}$. In addition, the monotone convergence curve could allow us to use Richardson's extrapolation in order to improve our best price estimates:

$$R = \frac{4C_0\left(\frac{dt}{n^2}, \frac{h}{n}\right) - C_0\left(\frac{4dt}{n^2}, \frac{2h}{n}\right)}{3}, \text{where } n = 10$$

However, our best price estimates were already around the upper bound of Monte Carlo true price. Moreover, given the ascending curve, Richardson's extrapolation inevitably results in higher estimates. Therefore, in this case the Richardson's extrapolation actually led to less precise prices, accurate to $10^{-1}$ or $10^{-2}$. The extrapolated values are reported in Table 1.

Table 1: Best price estimates

| Strike | Monte Carlo 95% CI | | Explicit Finite Difference Method | Richardson's Extrapolation | Geometric average put option |
|---|---|---|---|---|---|
| | Lower bound | Upper bound | | | |
| 95 | 3.5774 | 3.5789 | 3.5802 | 3.5823 | 3.9022 |
| 100 | 5.1052 | 5.1068 | 5.1069 | 5.1093 | 5.4867 |
| 105 | 6.9565 | 6.9583 | 6.9578 | 6.9603 | 7.3979 |

## Question 3

Turnbull and Wakeman (1991) provide with a closed-form solution for Geometric average Asian put option:

$$P = e^{-rT}KN\left(-\frac{\ln\frac{S_0}{K} + \left(r - \delta - \frac{\sigma^2}{2}\right)\frac{T}{2}}{\sigma\sqrt{\frac{T}{3}}}\right) - S_0 e^{-\left(\frac{r+\delta}{2}+\frac{\sigma^2}{12}\right)T} N\left(-\frac{\ln\frac{S_0}{K} + \left(\frac{r-\delta}{2} + \frac{\sigma^2}{12}\right)T}{\sigma\sqrt{\frac{T}{3}}}\right)$$

The prices are reported in the last column of Table 1. By definition, the geometric average of the stock prices is lower than the arithmetic average, because of the compounding effect. Therefore, the value of the geometric average put option is always greater than the value of its arithmetic average counterpart.

## Reference list

Kyriakou, I. (2020) 'Lecture 3. Variance Reduction'. *SMM313 Numerical Methods: Applications*. Cass Business School.

Turnbull, S. M., Wakeman, L. M. (1991) 'A Quick Algorithm for Pricing European Average Options', *The Journal of Financial and Quantitative Analysis*, 26(3), pp. 377–389. Available at: https://www.jstor.org/stable/2331213?seq=1. (Accessed: 2 April 2020).

Večeř, J. (2002) 'Unified Pricing of Asian Options'. Available at: http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.132.9508. (Accessed: 2 April 2020).